

Project Plan

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Motivation

Diffusion models [1] are very popular generative models that have displayed high quality samples in image, audio, video and graph generation. [2] [3] [4] [5] They learn a target data distribution through two processes. A forward process, that gradually adds noise to clean training data and then a reverse process that gradually moves from total noise to a clean generated sample. Generally, these generated samples will inherit any distributions that the training data had. So if the training data is biased or distributed somehow unfavorably or undesirably, then the generated samples will follow suit. Even if the training data is acceptable it can still be useful to control the distribution of the generation process. Examples include, generating 50% male and 50% female samples from a biased dataset, or generating samples of only a single color or variety. Whatever the case may be, controlling the generation process is often desirable.

Standard conditioning, which strictly conditions on a fixed observation y can be accomplished with methods such as Twisted Diffusion Sampling [6]. A more general method, Feynman-Kac steering [7], allows for arbitrary reward functions that can steer the model during inference toward user preferences such as a desired target distribution.

Jeffrey's conditioning [8] is a general technique for updating a joint probability distribution to satisfy some constraints. It uses the standard factorization of a joint probability distribution and then updates the factored marginal distribution to a new, desirable distribution and defining the new joint distribution accordingly. The goal of this thesis is to combine Jeffrey's rule with methods for guided sampling in order to steer the generated output of a diffusion model toward a desired distribution. This thesis will explore and evaluate this combined approach.

Background

Score-based diffusion models, currently the most prevalent paradigm of diffusion models, learn by estimating the score, which is the gradient of the log-density of the noise-perturbed data distribution. They can be described by stochastic differential equations[9]. Specifically, the diffusion process can be described by a forward SDE:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

and a reverse-time SDE:

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g^2(t)\nabla_{\mathbf{x}} \log p_t(\mathbf{x})]dt + g(t)d\bar{\mathbf{w}}$$

which yields a generative model of the data. Sampling is done by solving the reverse-time SDE using a trained neural network that has learned the score, $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

In order to steer the sampling toward a desired distribution, Jeffrey’s rule of conditioning can be used. It states that when we have some probabilistic model over d features, that is, we have a probabilistic model over $\mathcal{Z} = \mathcal{Z}_1 \times \mathcal{Z}_2 \times \dots \times \mathcal{Z}_d$ and we are unhappy with the marginal distribution of a subset of them, say $J \subset \{1, \dots, d\}$. We call this $p(z_J)$. Furthermore, we have an updated marginal, $p^*(z_J)$ that we are happy with. Taking $I = \mathcal{Z} \setminus J$ as the set of features we don’t want to change we can write the joint distribution, using the classic factorization, as:

$$p(z_I, z_J) = p(z_I|z_J)p(z_J)$$

The updated joint density is then given by:

$$p^*(z_I, z_J) = p(z_I|z_J)p^*(z_J)$$

Combining this, we obtain:

$$p^*(z_I, z_J) = \frac{p^*(z_J)}{p(z_J)}p(z_I, z_J)$$

The new distribution p^* is therefore an importance-weighted version of the original p . By using guidance methods such as TDS or Feynman-Kac steering and Jeffrey’s rule it is possible to steer the generated samples to follow a desired distribution.

Time Plan

Below is a time plan for the thesis, note that the process is inherently iterative and many parts will be done simultaneously, such as literature review and thesis writing. In addition, the process may evolve and change slightly as ideas or breakthroughs occur. This time plan should act as an outline to support the work of the thesis.

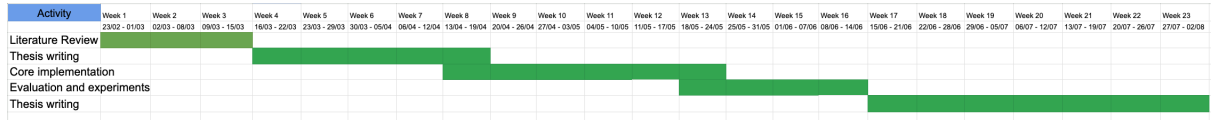


Figure 1: Time plan for thesis

Bibliography

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